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Specialty : Computer Systems

TITLE:

Development of a Secure and Intelligent Mobile System
for Psychological Support.

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Dedication:

To our beloved parents,

For their unconditional love, endless sacrifices, and unwavering faith in us. Every step of this journey has been made possible through your support and encouragement.

To our dear friends,

For the laughter, the shared struggles, and the cherished memories that made this academic journey meaningful. Your companionship and encouragement have been invaluable.

To all those who have supported us along the way,

Your kindness, guidance, and belief in us have left a lasting impact on our hearts and our success.

This work is dedicated to all of you, with deepest gratitude and love.

-OTMAN Kaouter-SALAH Aya-

Abstract

Mental health support is still out of reach for millions of people due to cost, availability, and stigma. Mind Space is an Android application that gives users a private space to reflect on their emotions through guided AI conversations. Sessions are automatically grouped into recurring emotional themes called Arcs, and users can generate insights on how these themes evolve over time. The project covers theoretical foundations, system modelling using RUP and UML, and a full implementation using Flutter, Supabase, and three AI services.

Keywords: mental health, mobile application, artificial intelligence, emotional reflection, Flutter, Supabase.

Résumé

Le soutien en santé mentale reste inaccessible pour des millions de personnes en raison du coût, de la disponibilité et de la stigmatisation. Mind Space est une application Android qui offre aux utilisateurs un espace privé pour réfléchir à leurs émotions à travers des conversations guidées par IA. Les sessions sont automatiquement regroupées en thèmes émotionnels récurrents appelés Arcs, avec la possibilité de générer des analyses de leur évolution. Le projet couvre les fondements théoriques, la modélisation avec RUP et UML, et une implémentation complète avec Flutter, Supabase et trois services d'IA.

Mots-clés : santé mentale, application mobile, intelligence artificielle, réflexion émotionnelle, Flutter, Supabase.

ملخص

يعاني كثير من الناس من مشاكل نفسية لكنهم لا يحصلون على الدعم الكافي بسبب التكلفة العالية وقلة المختصين تطبيق أندرويد يتيح للمستخدم تسجيل مشاعره والتحدث مع مساعد Mind Space. والخوف من نظرة المجتمع ، الذي يستمع ويطرح أسئلة ثم يقدم تأملاً في نهاية كل جلسة. مع الوقت، تتجمع Sage ذكاء اصطناعي اسمه ، ويمكن للمستخدم لاحقاً الاطلاع على كيفية تطور Arcs الجلسات تلقائياً حسب المواضيع المتكررة في ما يُسمى هذه المواضيع. المشروع يغطي الجانب النظري المتعلق بعلم النفس والذكاء الاصطناعي، ثم نمذجة النظام، وأخيراً Flutter و Supabase التطبيق الكامل باستخدام.

Flutter، Supabase، الصحة النفسية، تطبيق موبايل، ذكاء اصطناعي، التأمل العاطفي، :الكلمات المفتاحية

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List of Abbreviations

Abbreviation	Full Expression	Page
AI	Artificial Intelligence	1
API	Application Programming Interface	6
EFT	Emotionally Focused Therapy	5
GDPR	General Data Protection Regulation	6
LLM	Large Language Model	3
mHealth	Mobile Health	2
NLP	Natural Language Processing	3
RUP	Rational Unified Process	13
SSE	Server-Sent Events	26
TLS	Transport Layer Security	6
UI	User Interface	34
UML	Unified Modeling Language	13
WHO	World Health Organization	2

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General Introduction

A lot of people struggle with their mental health but never get proper support, therapy is expensive, General Introduction

A lot of people are not doing well mentally. Depression, anxiety, and chronic stress affect hundreds of millions worldwide, yet most of them never talk to a professional, not because they do not want to, but because therapy is expensive, not always nearby, and still carries a stigma that stops many people from asking for help.

Smartphones and AI opened up a different possibility. Natural Language Processing has reached a point where a machine can hold a conversation that actually feels human, not a replacement for therapy, but useful for someone who just needs to feel heard. The problem is that most existing applications in this space were not built carefully. They overwhelm users, reward engagement with guilt-inducing streaks, or quietly share sensitive emotional data with advertisers.

Mind Space is our response to these failures. It is an Android application where users can talk through what they are feeling with an AI companion called Sage, receive a structured reflection at the end of each session, and watch their conversations group over time into recurring emotional themes called Arcs. Privacy is built into the architecture from the start, no emotional data reaches third parties, and the crisis detection system always fails safely.

The project follows the Rational Unified Process methodology and is organised into three chapters:

- **Chapter 1** covers the psychological and technical background, closing with a formal SRS based on the IEEE 830 standard.
- **Chapter 2** presents the system architecture through UML diagrams, use cases, sequence diagrams, class model, and deployment diagram.
- **Chapter 3** covers the technologies used and a full walkthrough of the application interfaces.

CHAPTER 1

Presentation And Fundamental Concepts

1.1 Introduction

In recent years, digital mental health tools have gained a lot of attention. The idea of using a smartphone to support psychological wellbeing is not new anymore, but most existing solutions still have serious gaps — in terms of design, privacy, and actual usefulness. This chapter presents the theoretical and technical background that shaped the Mind Space project. It covers the psychological models we relied on, the role AI plays in this type of application, the privacy considerations we had to address, and the formal specification of the system requirements.

1.2 Context and Problematic

Mental health is considered by the WHO to be one of the most neglected areas of public health globally. Depression and anxiety affect hundreds of millions of people, yet the majority of them never receive any form of structured support. [1]

The barriers are not new: high cost, limited availability of professionals, long waiting times, and a social stigma that still prevents many people from seeking help openly. [2]

Smartphone applications have become one of the most discussed responses to this situation. Studies show that certain well-designed apps can reduce anxiety and depressive symptoms, particularly when users engage with them consistently. [2]

That said, the current market is full of products that do more harm than good. Streak-based tracking systems create anxiety when users miss a day. Mood logging interfaces with dozens of options cause what psychologists call decision fatigue. And several popular wellness applications have been found — through independent privacy audits — to share sensitive emotional data with advertising platforms without informing users. [9]

Mind Space was designed specifically to avoid these problems. The goal was to build something genuinely useful, private by design, and respectful of the user's emotional state.

1.3 Key Definitions

- **Mental Health:** A state of well-being in which a person can cope with life's ordinary pressures, work productively, and contribute to their community — as defined by the World Health Organization. [1]
- **mHealth:** The practice of medicine and public health through mobile devices. In this project, it refers to smartphone-based tools for emotional and psychological support.
- **Artificial Intelligence (AI):** Computer systems that simulate human reasoning — in this context, the language models used to generate responses in the journaling interface.
- **Natural Language Processing (NLP):** A subfield of AI focused on enabling machines to read, understand, and generate human language. [7]
- **Large Language Models (LLMs):** Deep learning models trained on large text corpora, capable of producing contextually relevant and fluent responses to open-ended prompts. [6]
- **Emotion Detection:** The computational process of identifying a person's affective state from written or spoken language. [7]
- **User-Declared vs. AI-Detected Emotion:** The User-Declared Emotion is what the user consciously selects. The AI-Detected Emotion is what the system infers from the content of their writing. The difference between the two is often more revealing than either alone.

1.4 Psychological foundations

1.4.1 Expressive Writing and Mental Health

Pennebaker's work established something that has since been confirmed in numerous studies: writing about emotionally difficult experiences — actually processing them rather than just describing them has measurable psychological benefits. [3]

This is the foundation that journaling applications build on. Mind Space takes it further by making the process conversational. Rather than facing a blank text field, the user is guided through the session by Sage, which asks questions and reflects back what it hears. This is closer in structure to how a supportive conversation actually works.

1.4.2 Emotion Classification Models

Two models from emotion psychology are particularly relevant here.

- **Ekman's Categorical Model:** Paul Ekman identified six basic emotions — Joy, Sadness, Anger, Fear, Disgust, and Surprise — that appear to be universal across human cultures. These form the basis of the emotion selection interface in Mind Space. [4]
- **Russell's Circumplex Model:** James Russell proposed mapping emotions onto two dimensions: valence (pleasant to unpleasant) and arousal (calm to activated). This gives a more nuanced picture of emotional states than a fixed list alone. [5]

We used Ekman's categories for the user-facing emotion selector because it is simple enough not to overwhelm people. Russell's dimensional thinking is more present in how the AI interprets the emotional texture of a conversation — not just what was named, but how it felt.

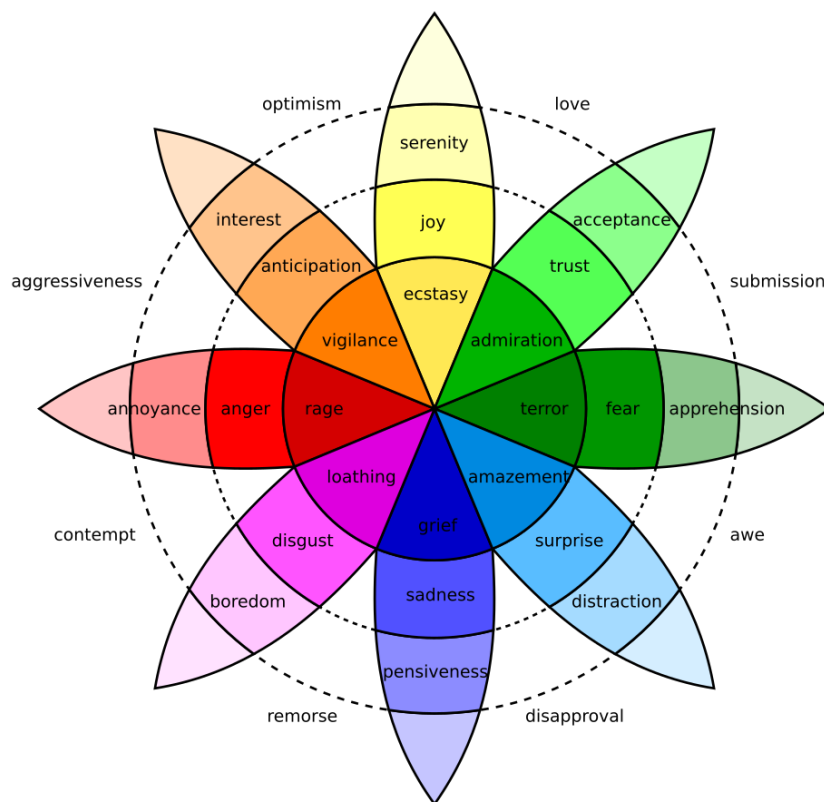


Figure 1- Visual representation of Plutchik's Wheel of Emotions[5].

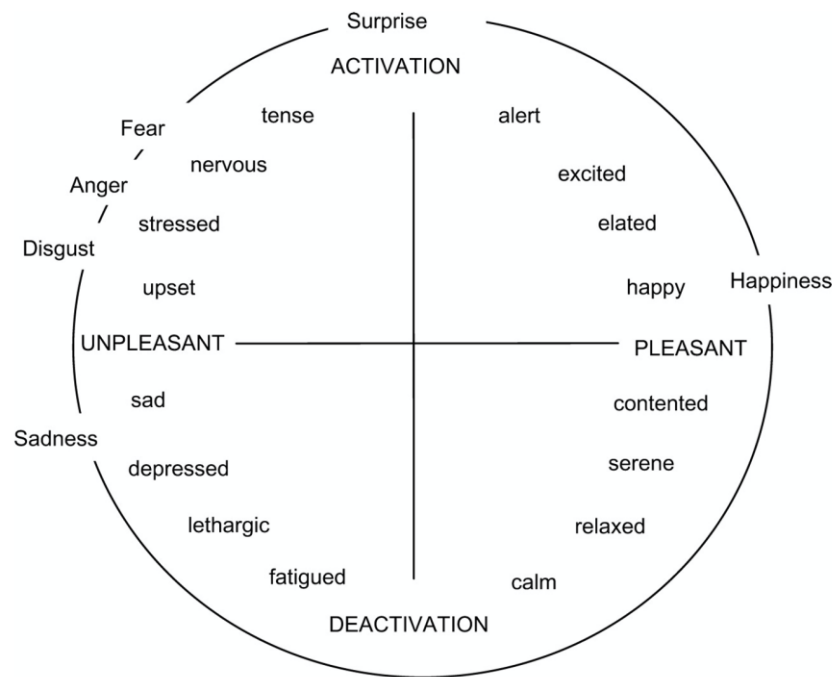


Figure 2- Russell's Circumplex Model of Affect [5].

1.5 Artificial Intelligence in Mental Health

1.5.1 Conversational Agents

Early chatbots for mental health were rule-based. They worked within narrow scripts and broke down as soon as users expressed themselves in unexpected ways. Large Language Models changed this considerably — they can sustain open conversations, pick up on emotional shifts in tone, and respond in a way that adapts to what is actually being said rather than what was anticipated. [6]

Sage, the AI companion in Mind Space, runs on this kind of model. It is not a therapist. It does not diagnose or advise. But it is designed to make the user feel genuinely heard, which is often the first thing people need.

1.5.2 Emotion Detection in Text

Detecting emotion from text is not about finding keywords. It is about reading patterns the words people choose, the topics they circle back to, the things they say and then qualify. A user can write "I'm doing okay" in a way that suggests anything but. [7]

In Mind Space, this is used to generate reflections that go a layer deeper than what the user explicitly stated. If the writing carries markers of anxiety under a surface claim of

calm, the reflection tries to name that. Carefully, and without telling the user what they should be feeling.

1.6 Data Privacy and security in mHealth

1.6.1 The Specific Risk of Psychological Data

Psychological data is qualitatively different from most other personal data. Financial records can be disputed. Passwords can be reset. A mental health history, once exposed, can affect how someone is treated at work, in personal relationships, and by insurance providers — in ways that are very difficult to undo. This makes the privacy architecture of a mental health application a core product decision, not a compliance checkbox.

1.6.2 Regulatory and Technical Standards

The standard requirements in this space are GDPR compliance, encryption of data at rest, and TLS for data in transit. [8]

The LINDDUN privacy threat modelling framework is useful for identifying specific risks in data flows — particularly how emotional data can be re-identified even when stored without a name attached. [9]

In Mind Space, no emotional data is routed through advertising or analytics services. The crisis safety classifier runs server-side via a dedicated Edge Function, and any content flagged as a crisis is neither stored in retrievable form nor forwarded to any external service.

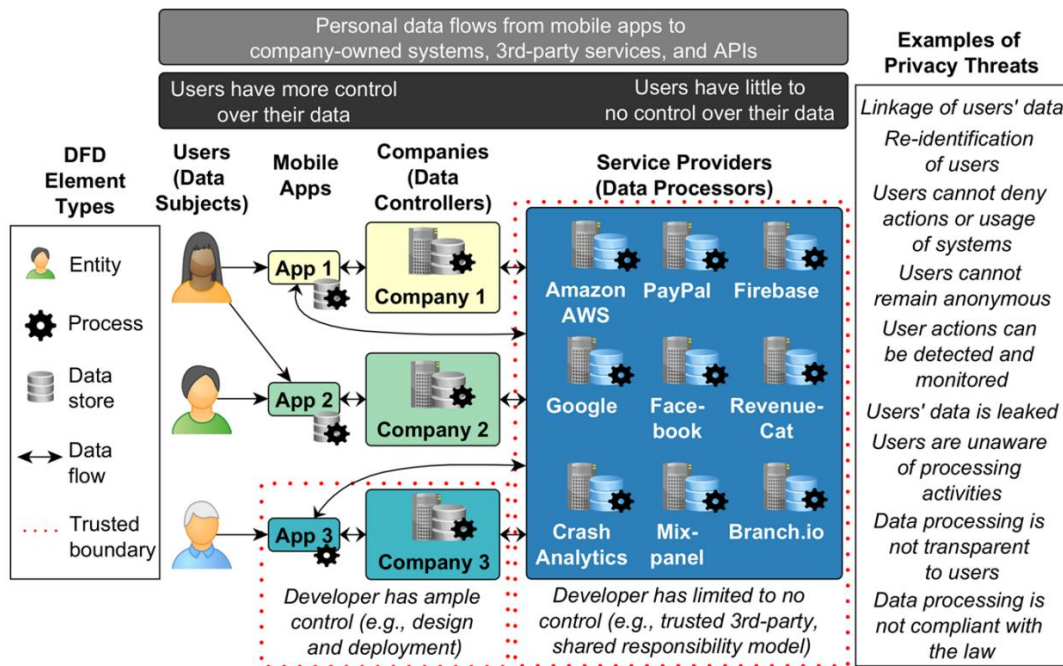


Figure 3- Simplified Data Flow Diagram (DFD) [9].

As illustrated in Figure 3, commercial applications frequently route user data through external analytics and marketing services, creating severe link ability and identifiability threats. To ensure user safety, next-generation mental health apps must rely heavily on local processing and zero-knowledge architectures, particularly when handling crisis detection (e.g., self-harm language). sensitivity of Psychological Data

The integration of AI in healthcare raises severe ethical and privacy concerns. Mental health data is among the most highly sensitive categories of personally identifiable information (PII). A security breach in an mHealth application can result in targeted exploitative advertising, social discrimination, and severe psychological distress for the user.

1.7 Analysis of Existing Solutions

Several applications currently occupy this space, and examining them is useful for understanding what Mind Space is responding to.

Wysa and Woebot are among the more respected conversational tools in this category. Both use structured therapeutic techniques and have some clinical backing. Their main limitation is that the conversation flows are relatively constrained — they handle expected inputs well but struggle when users go off-script.

Daylio is widely used for mood tracking. Its data visualisation is genuinely useful, but the emotion selection process is so detailed that it becomes a task in itself, which contradicts the goal of reducing friction.

The privacy dimension is where the market as a whole fails most consistently. Independent audits have found that several popular mental health applications share user data with third-party advertising networks — including Facebook — without making this explicit to users. [9]

Mind Space takes a different position on each of these points: the conversation is open-ended, the emotion interface is minimal, and no user data reaches external platforms.

1.8 Proposed Solution: Mind Space

MindSpace is a mobile app for emotional sense-making, not a clinical tool. The user selects an emotion from Ekman's six basic emotions [4], then enters a real-time conversation with Sage. After at least three exchanges, the system generates a structured reflection that names the emotional layer beneath what was said, along with a question to sit with [6]. Each reflection is embedded as a vector and compared to existing clusters if similarity exceeds 0.78, it joins an existing Arc; otherwise a new Arc is created, gradually mapping the user's recurring emotional themes [6][7]. A safety classifier screens every message before it reaches the model, and if crisis content is detected, the conversation pauses and emergency resources are shown [8][9].

1.9 Software Requirements Specification (SRS)

This section presents the formal requirements for Mind Space following the IEEE 830-1998 standard.

1.9.1 Object

To define the functional and non-functional requirements for a secure, AI-powered Android application for emotional self-reflection and psychological support.

1.9.2 Scope

Mind Space covers four functional areas:

- Psychologically grounded emotion logging based on validated models. [4][5]

- AI-powered conversational journaling to support expressive writing. [3][6]
- Analytical comparison of declared and detected emotion. [7]
- Privacy-first architecture with server-side crisis detection. [8][9]

1.9.3 Definitions

- **User-Declared Emotion:** The emotional state the user consciously selects from the interface.
- **AI-Detected Emotion:** The affective state inferred by the system from the user's written text.
- **Mood Sphere:** A visual component representing the logged emotion through standardised colour coding.
- **Arc:** A cluster of thematically related sessions grouped automatically by semantic similarity.
- **Unrestrictive Crisis Intervention:** A safety mechanism that presents emergency resources without closing the journaling interface.

1.9.4 References

(These correspond directly to the state-of-the-art literature established in Chapter 1).

- [1] WHO, "World Mental Health Report: Transforming Mental Health for All," 2022.
- [3] J. W. Pennebaker, "Writing about emotional experiences as a therapeutic process," 1997.
- [4] P. Ekman, "An argument for basic emotions," 1992.
- [5] J. A. Russell, "A circumplex model of affect," 1980.
- [6] T. Brown et al., "Language models are few-shot learners," 2020.
- [7] S. Poria et al., "A review of affective computing," 2017.
- [8] C. Nebeker et al., "Building the case for actionable ethics in digital health research," 2019.
- [9] L. H. Iwaya et al., "On the privacy of mental health apps," 2023.

1.9.5 General Description

A. Operating Environment

Android application built with Flutter (Dart). Backend: Supabase with PostgreSQL, Row-Level Security, and serverless Edge Functions (Deno/TypeScript). AI services: Groq for conversation and safety classification, OpenRouter for Arc insight generation, OpenAI for semantic embeddings. No user data passes through advertising or analytics services. [9]

B. End-User Characteristics

- Young adults and university students experiencing emotional stress or academic pressure.
- Comfortable with smartphones and AI-based interfaces.
- Seeking genuine self-reflection without gamification or clinical overreach.

C. System Constraints

- No feature is accessible without prior authentication.
- The AI conversation requires an active internet connection. [6]
- Data in transit is protected by TLS 1.3; database access enforced by Row-Level Security. [8]
- No emotional data is shared with third-party advertising or analytics services. [9]

1.9.6 Specific Requirements

A. Functional Requirements

- **FR-01 (Secure Authentication):** Users sign in via Google OAuth or email magic link. A Postgres trigger creates the user profile automatically on first sign-in.
- **FR-02 (Categorical Emotion Logging):** At the start of each session, users select their emotional state from a set of core emotions based on Ekman's model. [4]
- **FR-03 (Dimensional Nuance):** Users can optionally add sub-emotions or contextual tags to give the AI richer input data. [5]
- **FR-04 (Conversational Journaling):** Sage responds to user messages with empathetic, non-diagnostic prompts. Responses stream token by token. [3][6]

- **FR-05 (Affective Computation):** After at least three exchanges, the system generates an EFT-informed reflection: the detected emotional layer, a question to sit with, and a normalising perspective. [7]
- **FR-06 (Arc Auto-Assignment):** Each reflection is embedded and compared to Arc centroids using cosine similarity. A score above 0.78 assigns it to the closest Arc; below that, a new Arc is created and named by the language model. [6]
- **FR-07 (Perception Comparison):** An analytical view shows how the user's declared emotion compares to what the AI detected — presented without judgement or correction. [7]
- **FR-08 (Pattern Visualisation):** The Analysis screen displays a 14-week session heatmap and emotional breakdowns per Arc.
- **FR-09 (Data Portability):** Users can export their full session history and AI analyses as a JSON file.
- **FR-10 (Crisis Intervention):** Every message passes through a safety classifier before reaching the main LLM. If crisis content is detected, the chat is paused and a modal shows emergency resources including Algeria-specific contacts. The classifier fails closed — if unavailable, the chat is blocked. [8][9]

B. Non-Functional Requirements

- **NFR-01 (Performance):** The emotion logging flow completes in under 20 seconds.
- **NFR-02 (Latency):** The first AI response token appears within 3 seconds under normal network conditions.
- **NFR-03 (Security):** TLS 1.3 for data in transit; Supabase Row-Level Security at the database level. [9]
- **NFR-04 (Privacy):** Crisis-flagged content is not stored or forwarded. No emotional data reaches third-party platforms. [8][9]
- **NFR-05 (Reliability):** Pipeline failures are surfaced to the user with a clear retry option — no silent errors.
- **NFR-06 (Maintainability):** The Sage prompt, safety prompt, and similarity threshold are Edge Function variables — updatable without redeploying the application.

1.10 Conclusion

This chapter laid the theoretical and technical foundation of the project — from the psychological models behind emotion tracking to the formal system requirements. The analysis of existing solutions helped clarify what Mind Space needed to do differently, particularly in terms of privacy and user experience.

The next chapter moves into the modelling phase, where these requirements are translated into system diagrams using the RUP method.

CHAPTER 2

Modelling

2.1 Introduction

This chapter presents the modelling of the MindSpace application. It begins with a brief overview of the UML language and the RUP development methodology used to guide the design process, then moves into the core of the chapter: the use case diagram, class diagram, and sequence diagrams that define the structure and behaviour of the system.

2.2 UML —Unified Modeling Language

UML (Unified Modeling Language) is a standardised visual language for specifying and documenting software systems, maintained as an international standard by the Object Management Group (Booch et al., 2005). For this project, three diagram types were used: the Use Case Diagram to define system functionality per actor, the Class Diagram to describe the data structure and object relationships, and Sequence Diagrams to capture the dynamic behaviour of the most critical system flows .

2.3 RUP Method

RUP is an iterative software development methodology structured around four phases: Inception, Elaboration, Construction, and Transition (Kruchten, 2003). It was chosen for MindSpace because it suits projects where requirements evolve during development. The Inception phase established the project concept. The Elaboration phase produced the architecture and the UML models in this chapter. The Construction phase covers feature implementation, and the Transition phase covers testing, demo preparation, and defence.

2.4 Diagrams

In our project, we defined the following diagrams: use case, class, sequence and deployment diagrams.

2.4.1 Use Case Diagram

A use case diagram is a visual representation that identifies the interactions between external users (actors) and a system to achieve specific goals.

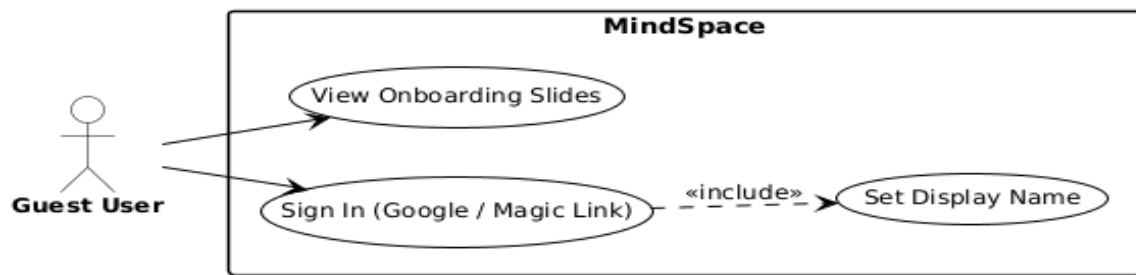


Figure 4 -Guest User

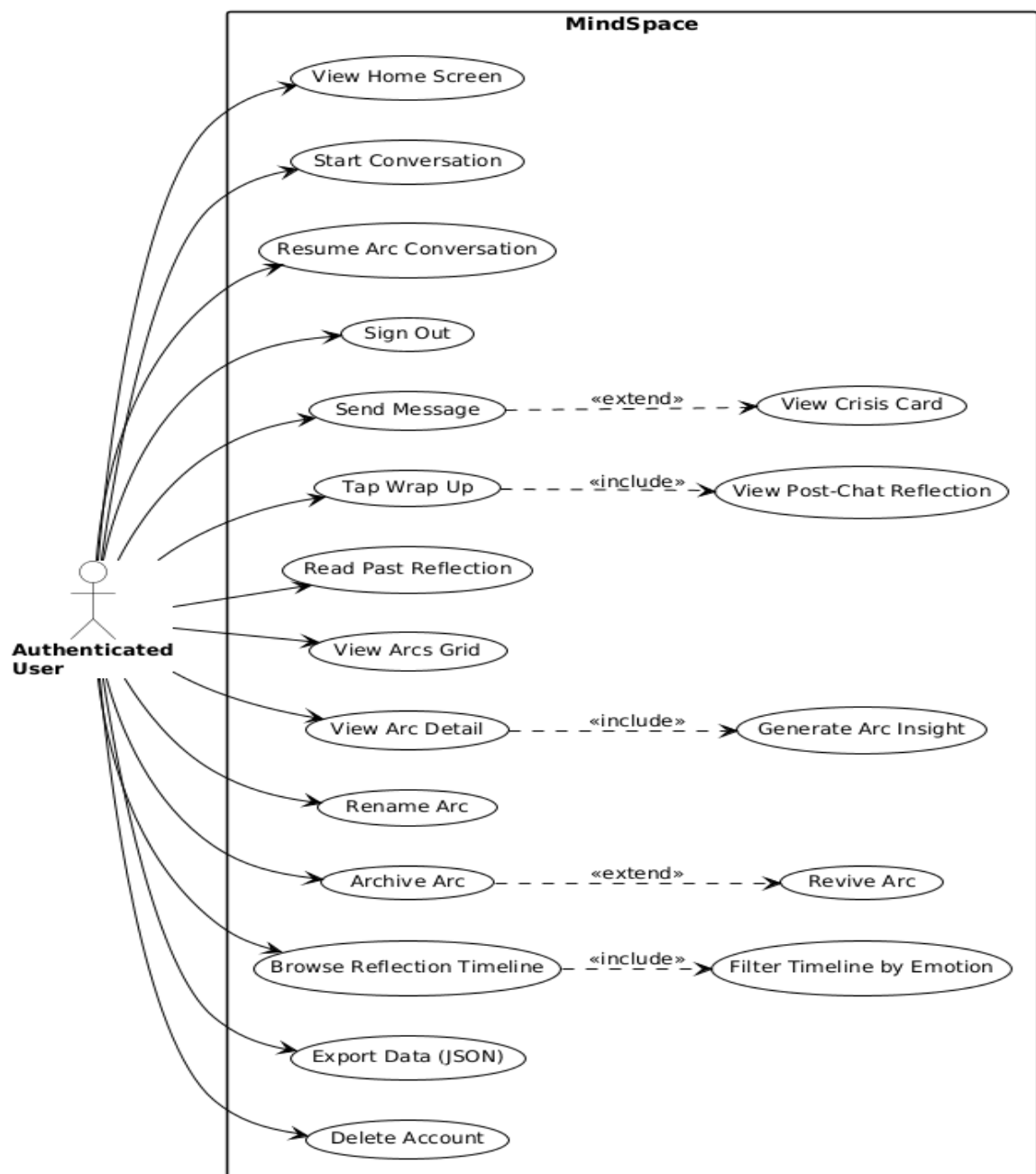


Figure 5 - Authenticated User

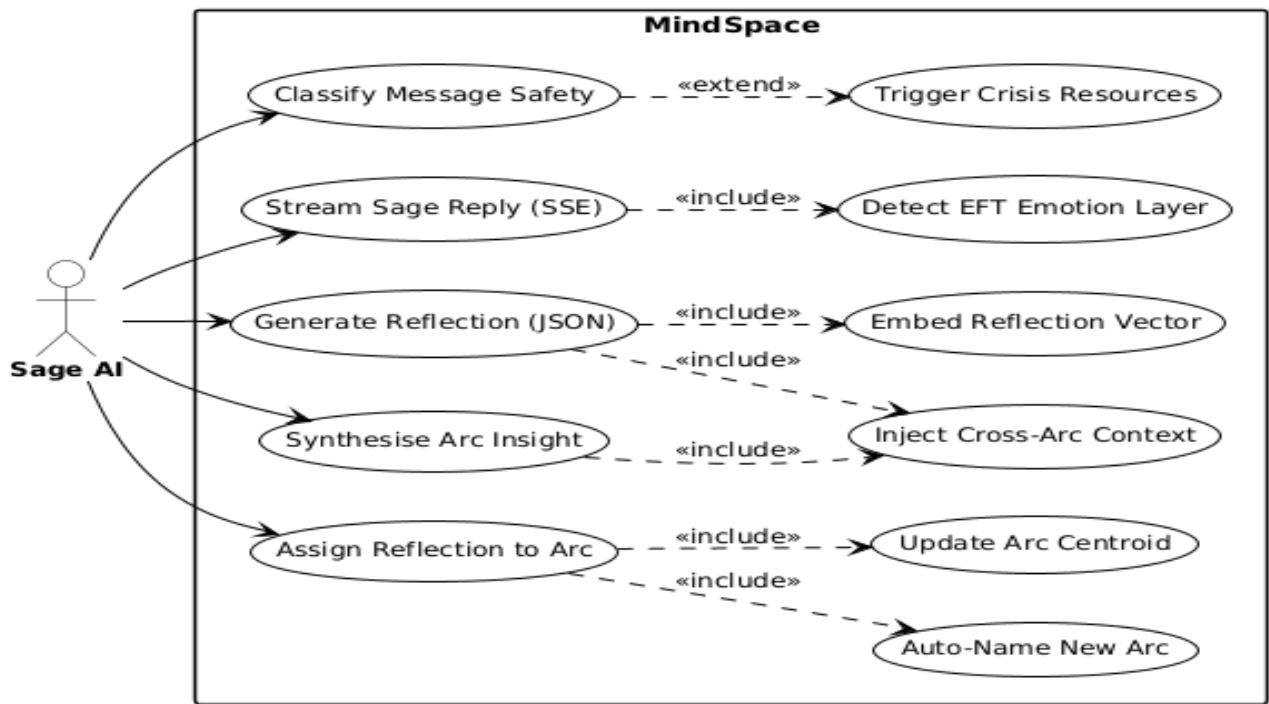


Figure 6- Sage AI

- **Explication**

The MindSpace use case diagram covers three principal actors. The **Guest User** in **Figure 4** handles entry-level interactions: viewing onboarding slides, signing in, and setting a display name. The **Authenticated User** in **Figure 5** covers the full app experience: starting and resuming conversations, sending messages, wrapping up sessions to generate reflections, managing and browsing Arcs, generating Arc insights, and handling account settings. The **Sage AI** actor in **Figure 6** manages all backend pipelines: safety classification, streaming replies with EFT emotion detection, generating and embedding reflections, assigning reflections to Arcs, and synthesising macro Arc insights.

2.4.2 Class Diagram

A class diagram is a static diagram that shows the structure of the system by representing the main classes, their attributes, their methods, and the relationships between them. It basically describes how the different parts of the application are organised and how they communicate with each other.

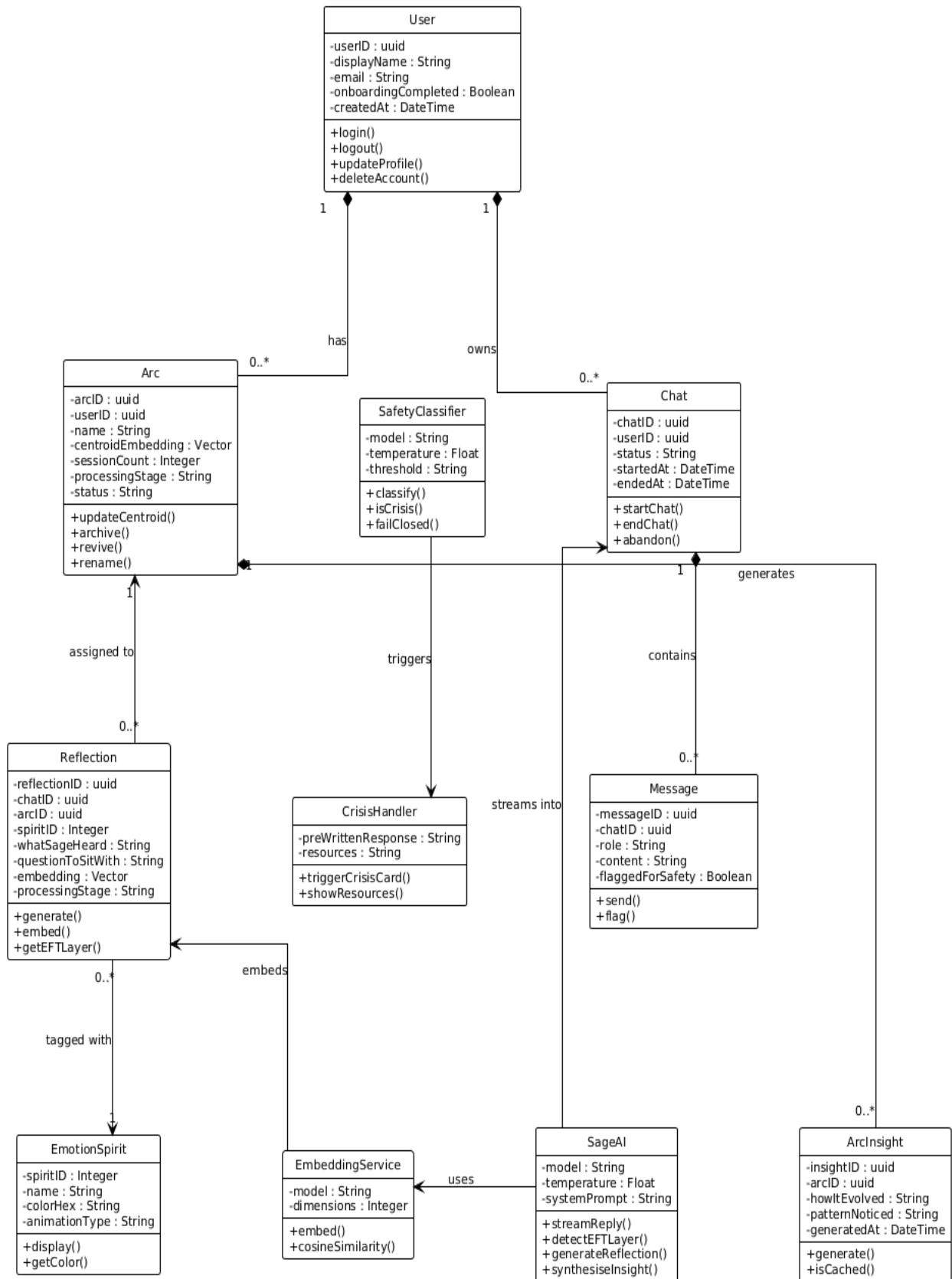


Figure 7- Class Diagram

- **Explication**

Figure 7 shows the class diagram of a reflective AI chat app. At the top, User owns both Chat sessions and Arc objects (emotional themes). Each Chat contains Messages that get monitored by a Safety Classifier — if a crisis is detected, it fires the Crisis Handler. Chats also generate Reflections (AI summaries via Sage AI), which get embedded as vectors by Embedding Service and tagged with Emotion Spirit objects for visual emotional indicators. Reflections are assigned to an Arc, and each Arc can produce an Arc Insight that tracks patterns over time.

2.4.3 Sequence Diagram

A sequence diagram is a UML behavioral diagram that represents how objects interact with each other over time to accomplish a specific functionality. It shows the order of messages exchanged between objects, emphasizing the chronological sequence of interactions from the start of a scenario to its completion.

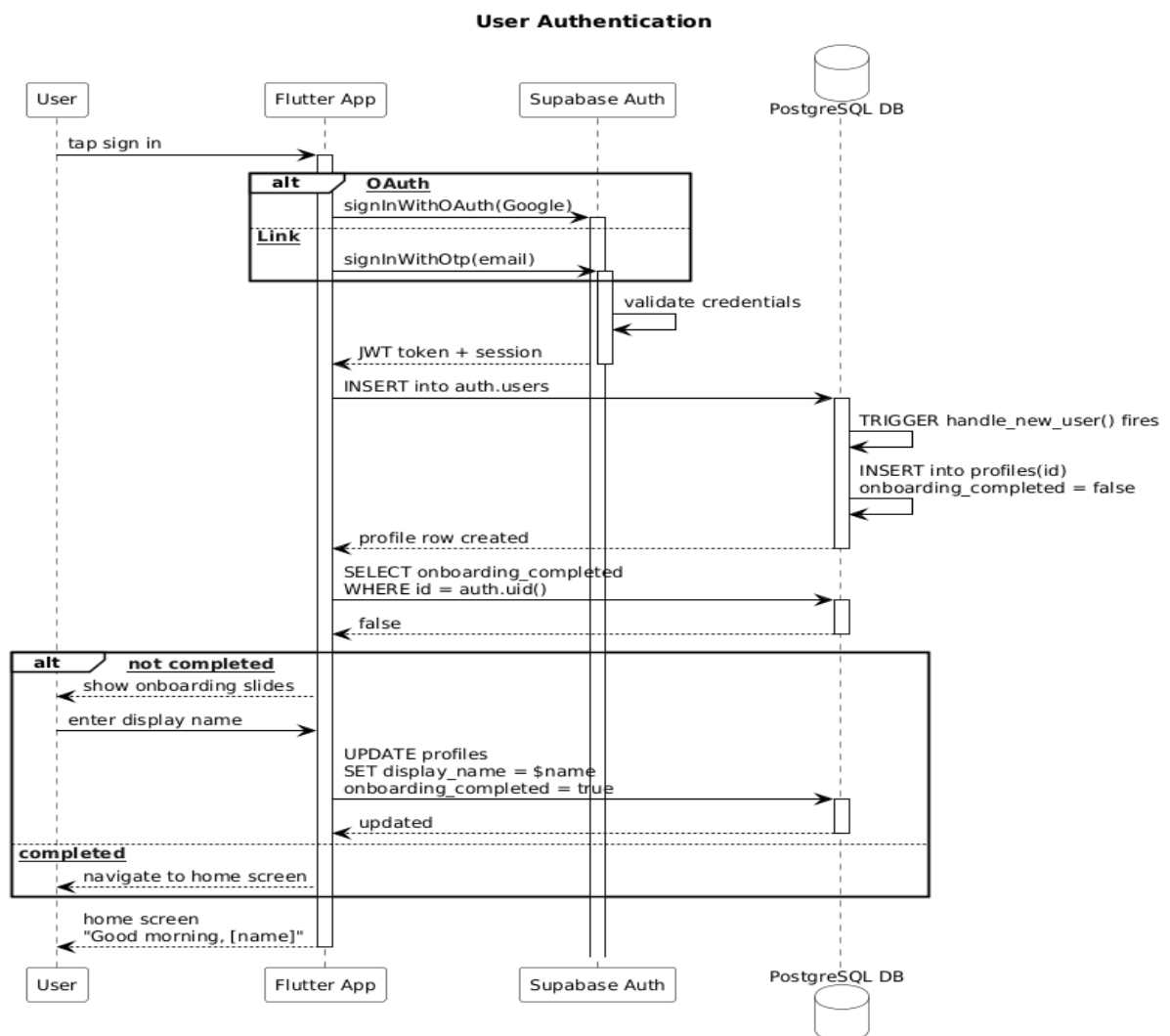


Figure 8 -Sequence Diagram of User Authentication

Table 1 -Textual Description of User Authentication

Use Case: User Authentication			
Objective	Authenticate the user and create their profile		
Primary actor	User		
Precondition	App is installed. User has a Google account or a valid email address.		
Post condition	User is authenticated. Profile row exists in DB. Display name is set. Onboarding is completed.		
Version	1.0		
	Nominal Scenario	Alternative Scenario	Error Scenarios
	1. User opens the app and taps Sign In 2. App sends request to Supabase Auth 3. Supabase Auth validates and returns JWT token 4. DB trigger handle_new_user() fires and inserts profile row 5. App checks onboarding_completed from DB 6. App navigates to onboarding slides 7. User enters display name 8. App updates profile: display_name and onboarding_completed = true 9. App navigates to home screen	A1 - Via Google: User authenticates through Google OAuth A2 - Via magic link: User enters email and clicks magic link A3 - Onboarding done: App navigates directly to home screen	E1 - Invalid credentials: Auth fails, app shows error and prompts retry E2 - Trigger fails: handle_new_user() does not fire, app retries profile read

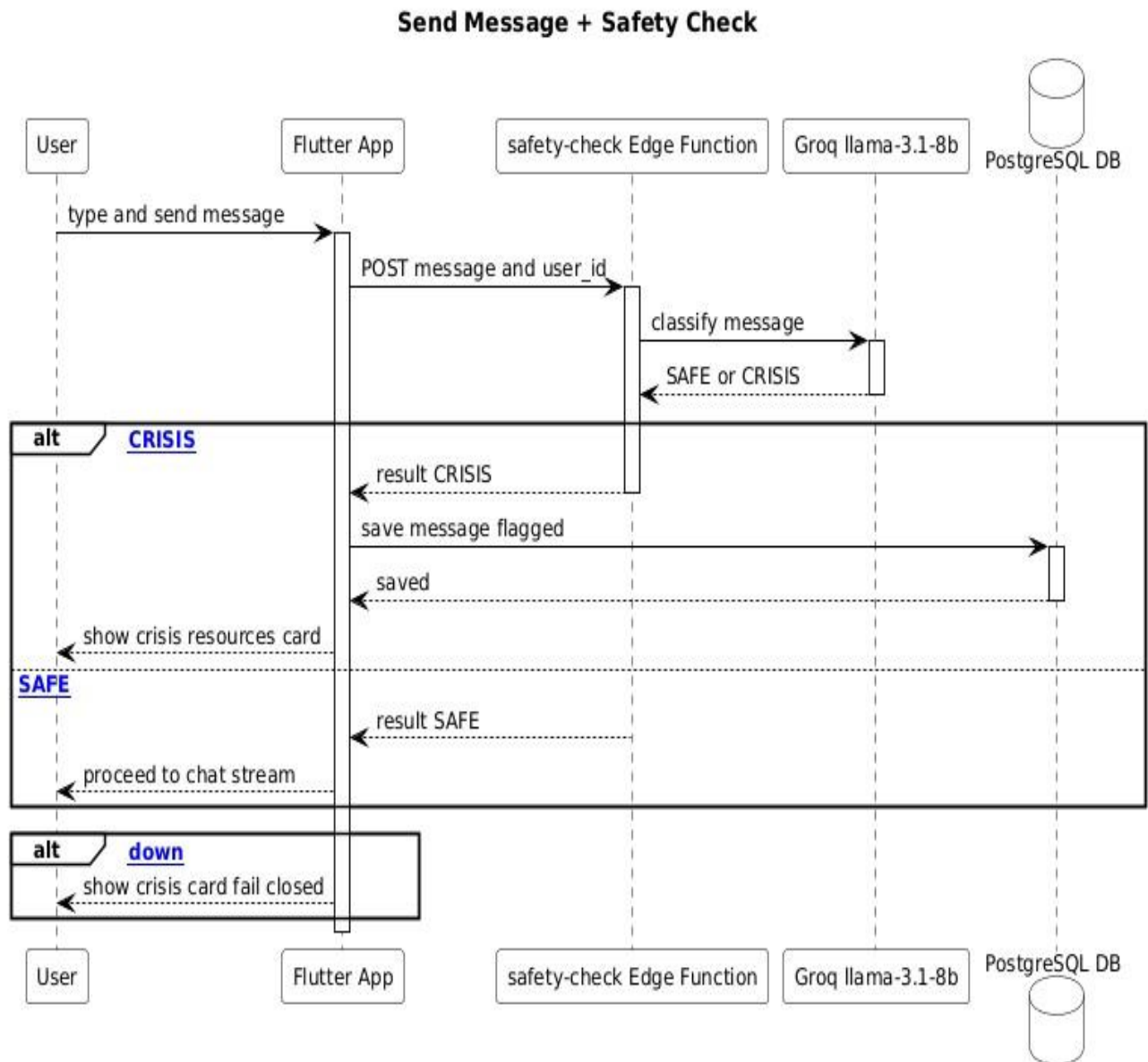


Figure 9 -Sequence Diagram of Send Message & Safety Check

Table 2 -Textual Description of Send Message & Safety Check

Use Case: Send Message + Safety Check	
Objective	Classify user message for safety before processing
Primary actor	User

Precondition	User is authenticated. Active chat session is open.		
Post condition	Message is classified. If SAFE chat proceeds. If CRISIS resources card is shown and message is flagged.		
Version	1.0		
	Nominal Scenario	Alternative Scenario	Error Scenarios
	1. User types and sends a message 2. App sends message to safety-check Edge Function 3. Edge Function calls Groq llama-3.1-8b at temperature 0.1 4. Groq returns SAFE or CRISIS 5. If SAFE app proceeds to chat-stream 6. Message saved with flagged_for_safety = false	A1 - CRISIS detected: App shows hardcoded response and Algeria crisis resources (SAMU 15), message saved with flagged_for_safety = true A2 - User taps I am safe: User dismisses crisis card and chat continues	E1 - API down: Safety check fails, system fails closed, crisis card shown regardless E2 - Timeout: Request to Groq times out, system treats as CRISIS

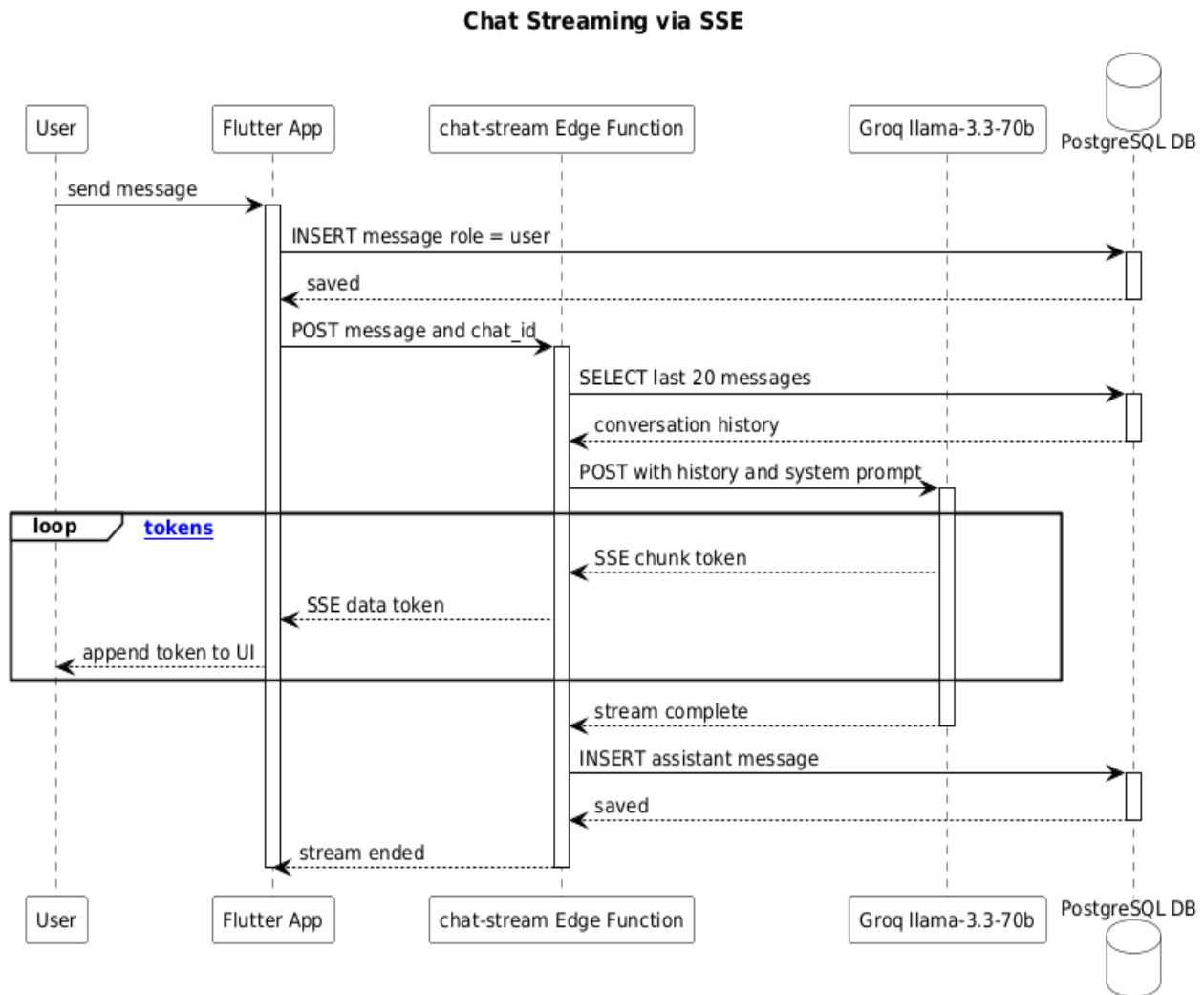


Figure 10 -Sequence Diagram of Chat Streaming via SSE

Table 3-Textual Description of Chat Streaming via SSE

Use Case: Chat Streaming via SSE	
Objective	Stream Sage AI reply token by token to the user via Server-Sent Events
Primary actor	User
Precondition	User is authenticated. Message has passed safety check.
Post condition	Sage reply is fully streamed. Both user and assistant messages are saved to DB.

Version	1.0		
	Nominal Scenario	Alternative Scenario	Error Scenarios
	1. App saves user message to DB with role = user 2. App sends POST to chat-stream Edge Function 3. Edge Function fetches last 20 messages from DB 4. Edge Function calls Groq llama-3.3-70b with streaming = true and temperature = 0.7 5. Groq streams reply tokens via SSE 6. Each token is forwarded to Flutter client 7. Flutter appends each token to UI in real time 8. Stream completes, full assistant message saved to DB	A1 - User closes app mid-stream: Partial message is lost, no partial message saved to DB A2 - After 4 user messages: Wrap Up pill appears, user can end the session	E1 - Groq API down: Edge Function returns error, app shows offline banner E2 - Stream interrupted: Connection drops mid-stream, token buffer discarded, user sees offline banner

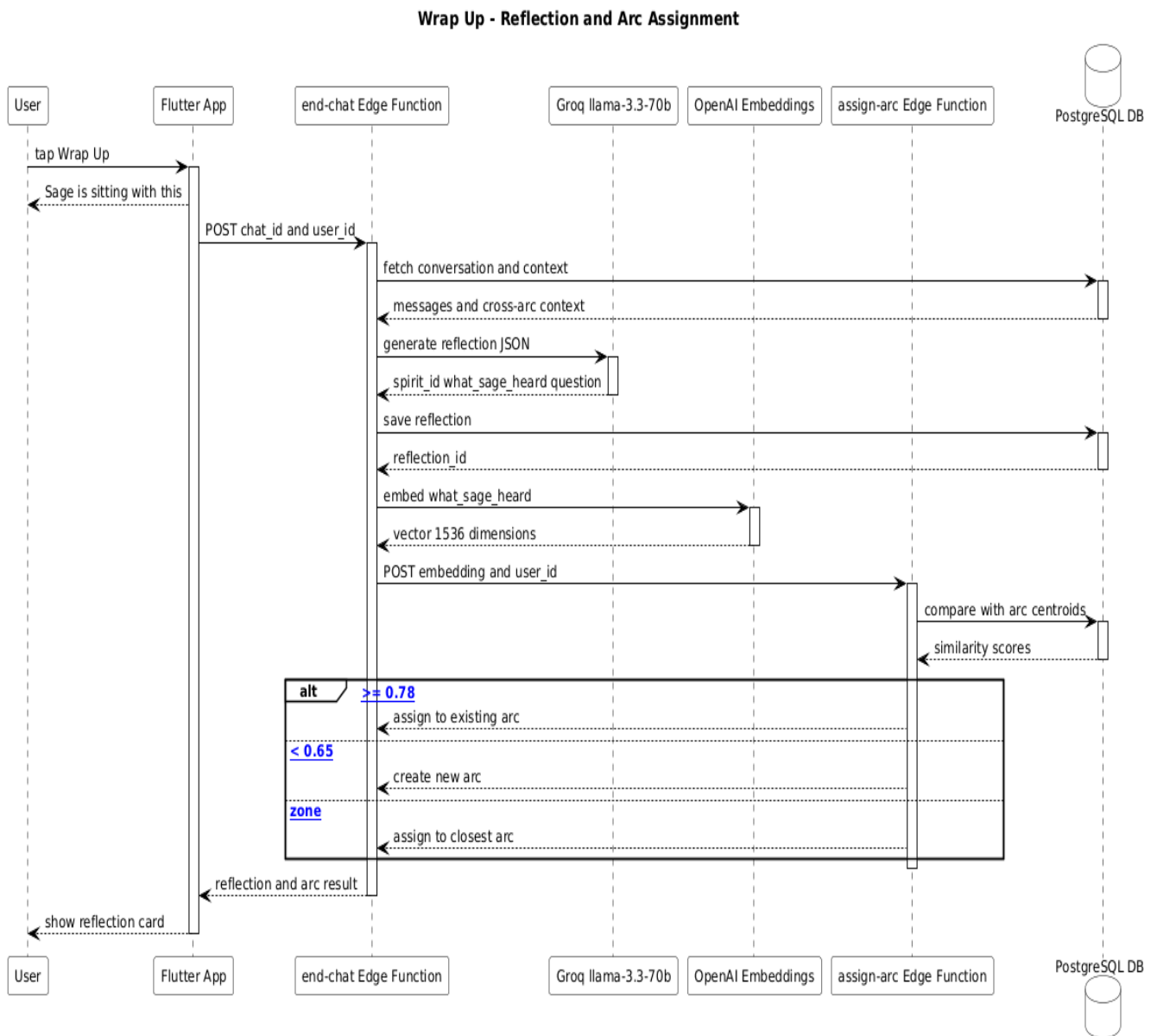


Figure 11 -Sequence Diagram of Wrap Up - Reflection and Arc Assignment

Table 4 -Textual Description of Wrap Up - Reflection and Arc Assignment

Use Case: Wrap Up - Reflection and Arc Assignment	
Objective	Generate EFT-layered reflection and assign it to an Arc after chat ends
Primary actor	User

Precondition	User is authenticated. Chat session has at least one exchange. User taps Wrap Up.		
Post condition	Reflection is generated and saved. Embedding vector is computed. Reflection is assigned to existing or new Arc. Reflection card is shown.		
Version	1.0		
	Nominal Scenario	Alternative Scenario	Error Scenarios
	1. User taps Wrap Up 2. App shows loading state 3. end-chat fetches conversation and cross-arc context from DB 4. Groq generates reflection JSON with spirit_id, what_sage_heard and question 5. Reflection saved to DB 6. OpenAI embeds what_sage_heard into 1536-dimension vector 7. assign-arc compares vector with arc centroids 8. Reflection assigned to arc 9. App shows reflection card to user	A1 - Existing arc assigned: Similarity ≥ 0.78 , centroid updated, session_count incremented A2 - New arc created: Similarity < 0.65 , new arc created with auto-generated name A3 - Soft zone: Similarity 0.65 to 0.78, assigned to closest arc	E1 - Reflection fails: App shows retry or skip button E2 - Embedding fails: Reflection saved with needs_arc_assignment = true for retry

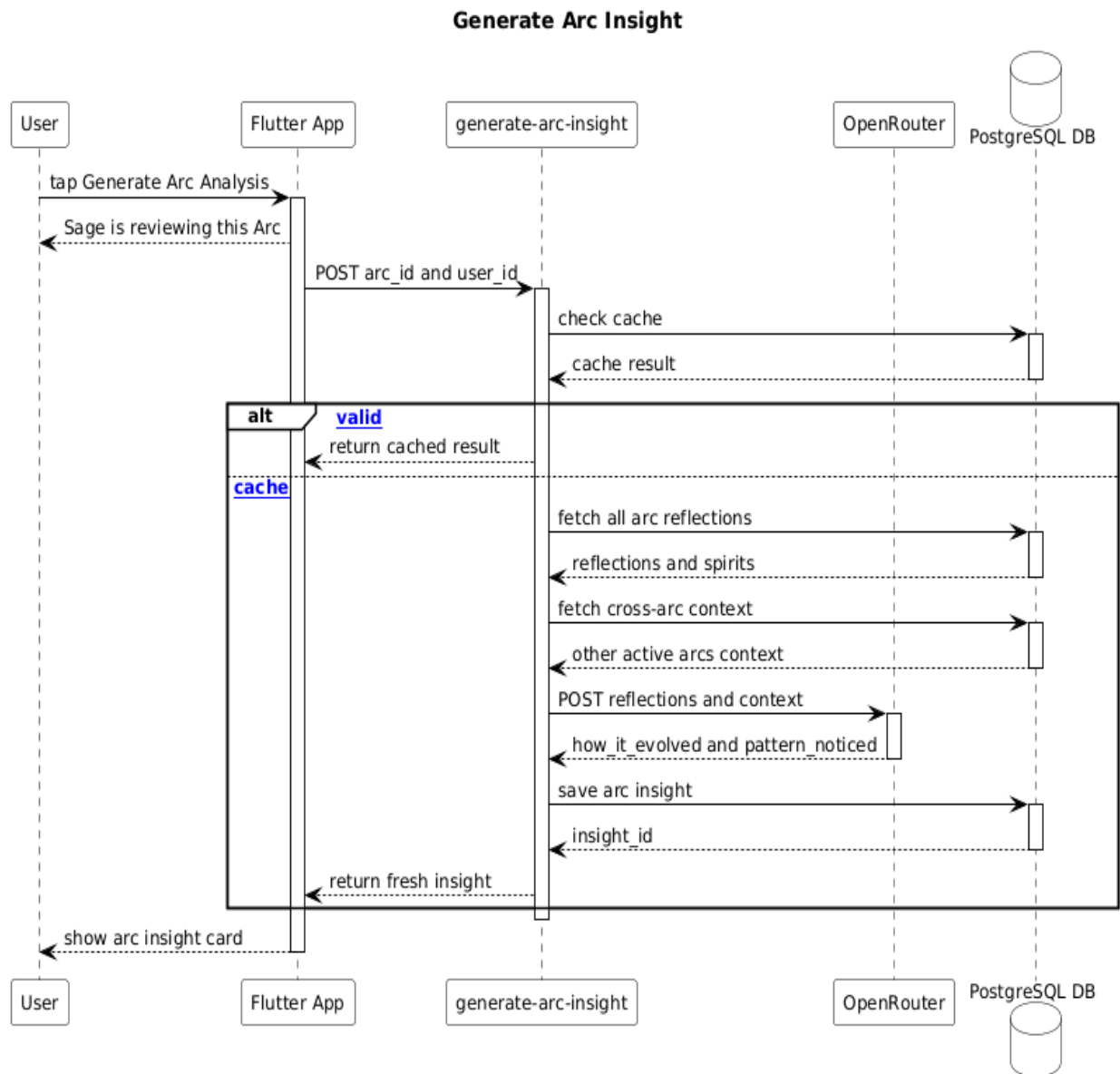


Figure 12 -Sequence Diagram of Generate Arc Insight

Table 5-Textual Description of Generate Arc Insight

Use Case: Generate Arc Insight	
Objective	Generate a macro insight for an Arc with 3 or more sessions
Primary actor	User

Precondition	User is authenticated. Arc has at least 3 sessions. User taps Generate Arc Analysis.		
Post condition	Arc insight is generated and saved. User sees how_it_evolved and pattern_noticed. Result cached for 7 days.		
Version	1.0		
	Nominal Scenario	Alternative Scenario	Error Scenarios
	1. User taps Generate Arc Analysis 2. App shows Sage is reviewing this Arc 3. generate-arc-insight Edge Function checks cache in DB 4. No cache found 5. Edge Function fetches all arc reflections chronologically 6. Edge Function fetches cross-arc context from other active arcs 7. OpenRouter generates how_it_evolved and pattern_noticed at temperature 0.3 8. Insight saved to DB with session_count_at_generation 9. App shows arc insight card	A1 - Cache valid: Insight generated less than 7 days ago and session_count unchanged, cached result returned immediately A2 - New session added: session_count changed since last generation, fresh insight generated	E1 - OpenRouter API down: Edge Function returns error, app shows retry button E2 - Rate limit reached: 200 RPD exceeded, app suggests trying again later

2.4.4 Deployment diagram

A deployment diagram shows how a software system is physically distributed across hardware and infrastructure nodes. It describes which components run where and how they communicate with each other. In MindSpace it shows the Flutter app on the Android device, the Edge Functions and database on Supabase Cloud, and the external AI APIs communicating over HTTPS.

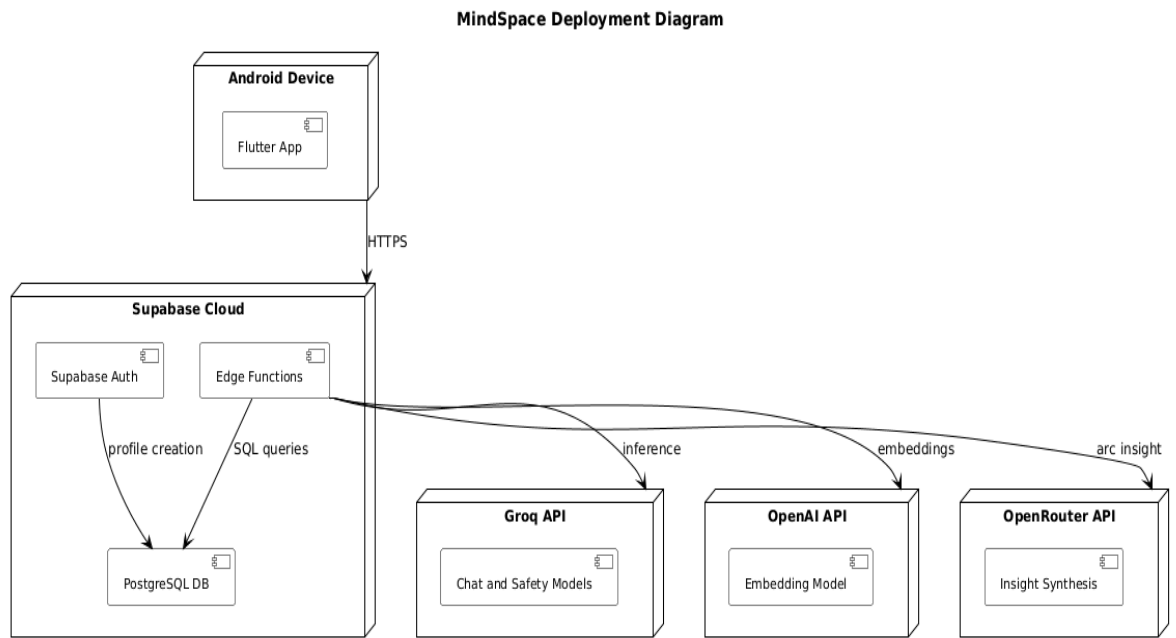


Figure 13 - Deployment diagram

- **Explication**

Figure 13 illustrates the deployment diagram of the MindSpace application. The Flutter App runs on an Android Device and communicates with the backend via HTTPS. All requests land on Supabase Cloud, which handles authentication through Supabase Auth and business logic through Edge Functions, both storing and retrieving data from a PostgreSQL DB. The Edge Functions also connect to three external APIs: Groq API for chat and safety model inference, OpenAI API for generating text embeddings, and OpenRouter API for arc insight synthesis.

2.5 Conclusion

Following the RUP methodology, this chapter covered the design of MindSpace through UML models including use case, sequence, class and deployment diagrams, addressing all key features of the application. The next chapter will focus on the construction phase, covering the implementation and testing of the application.

CHAPTER 3

Implementation of Mind Space

3.1 Introduction

This chapter presents the implementation of Mind Space, an Android application for emotional reflection powered by AI. It covers the tools and technologies used, the system architecture, and a presentation of the main application interfaces.

3.2 Technologies and Tools Used

The development of Mind Space relied on a set of carefully chosen technologies. The following presents the most important ones:

Flutter : An open-source UI framework developed by Google for building native Android and iOS applications from a single Dart codebase. It was used for the entire mobile frontend of Mind Space, along with the Riverpod library for state management and go_router for navigation.[10]

Supabase : An open-source backend platform built on PostgreSQL. It provides authentication, a relational database with Row-Level Security, serverless Edge Functions, and file storage — all on a free tier. Mind Space uses Supabase as its complete backend infrastructure.[11]

PostgreSQL + pgvector : The relational database used to store all application data. The pgvector extension enables native vector similarity search, which is essential for the Arc auto-assignment algorithm that clusters reflections by semantic closeness.[11]

Groq API : A high-speed inference service used to run two language models: llama-3.3-70b-versatile for the streaming Sage conversation and reflection generation, and llama-3.1-8b-instant for the pre-message safety classifier. Both models are available on the free tier.[6]

OpenRouter — DeepSeek-R1 : Used exclusively for Arc macro-insight generation. DeepSeek-R1 offers strong long-context reasoning, making it well suited for synthesising an emotional narrative across multiple sessions. Its free tier (200 requests/day) is sufficient for this use case.[6]

OpenAI text-embedding-3-small : Converts each reflection into a 1 536-dimensional semantic vector. These vectors are stored in PostgreSQL via pgvector and compared using cosine similarity to automatically assign reflections to the correct Arc.

Figma : Used for UI/UX design and prototyping. All color tokens, typography rules, and component specifications were defined in Figma before implementation.[10]

Visual Studio Code : The main code editor used throughout the project, with the Flutter and Dart extensions for debugging and hot-reload development.[10]

3.3 Presentation of the Application

The following sections present the main screens of Mind Space in the order a user encounters them.

- **Onboarding interface :** introduce the app concept, the Arc system, and the privacy guarantee. Onboarding is shown once on first launch only

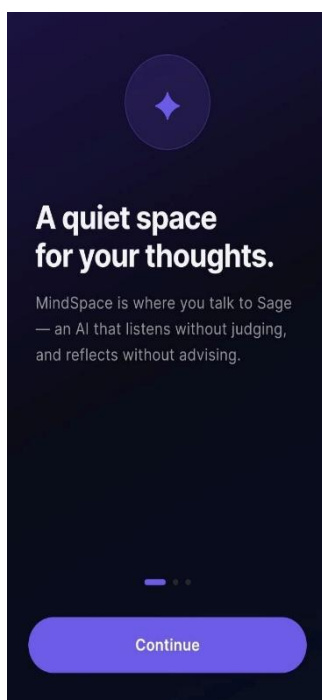


Figure 14 - Onboarding — Slide 1

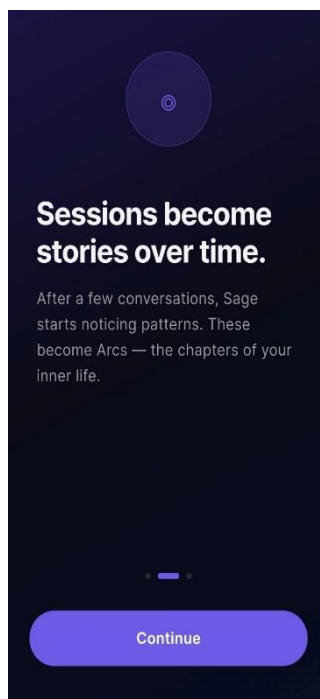


Figure 15 - Onboarding —Slide 2

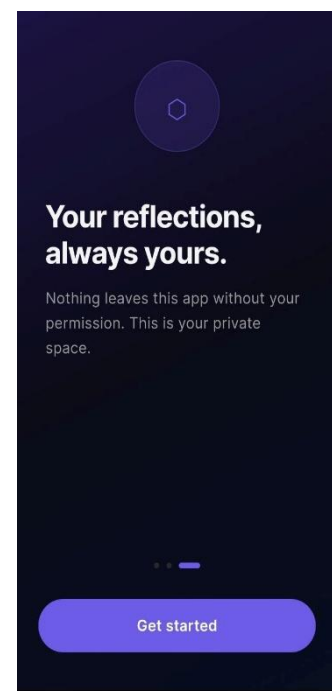


Figure 16 - Onboarding — Slide 3

- **Authentication:** The user signs in via Google OAuth or email magic link no password required. A database trigger automatically creates the user profile on first sign-in.

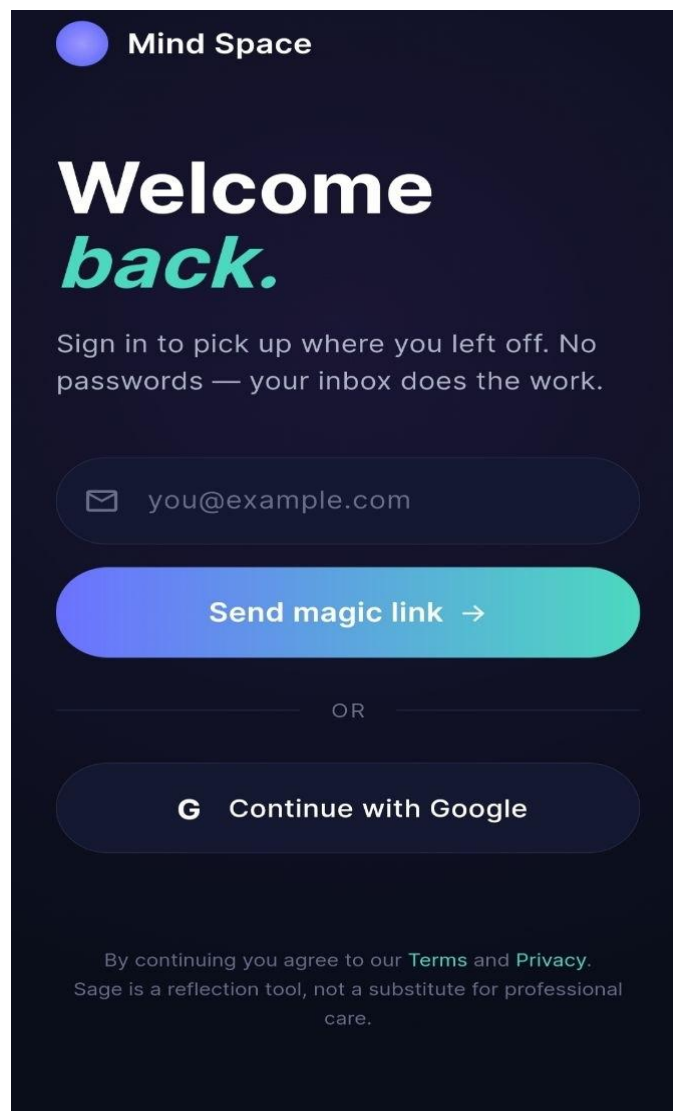


Figure 17 - Authentication Screen

- **Home and Chat Screens :** The Home screen shows active Arc cards and a personalised greeting. The Chat screen hosts the streaming conversation with Sage; after four messages a Wrap Up button appears to end the session and generate a reflection.

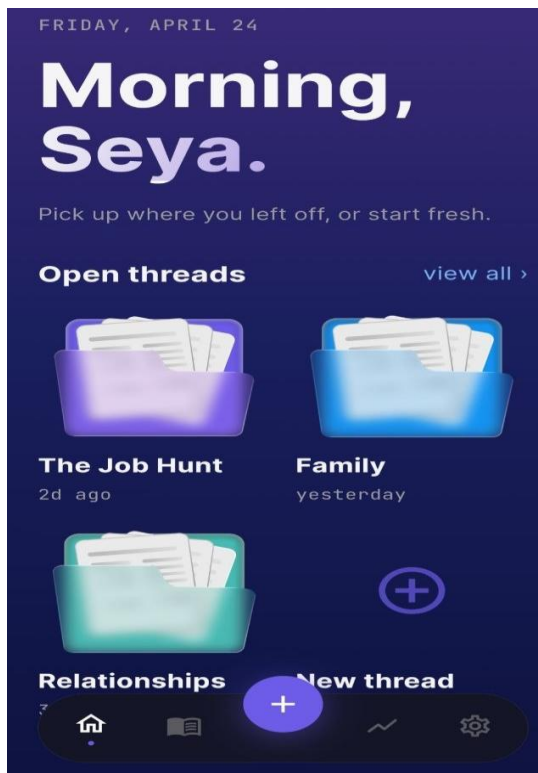


Figure 18 - Home Screen — Open Threads

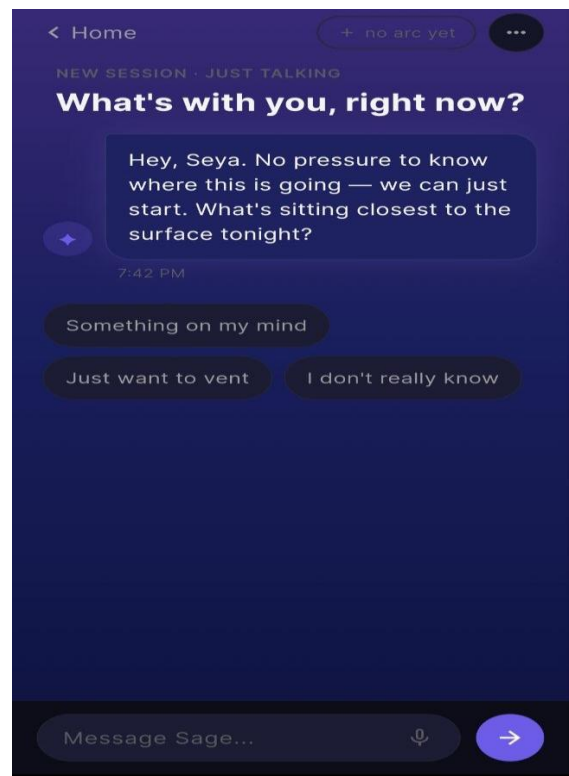


Figure 19 - Conversation with Sage

- **History — Timeline and Arcs Grid :** The History tab offers two views: a reverse-chronological Timeline of reflections filterable by emotion, and an Arcs Grid showing all Arc cards with their dominant emotion breakdown.

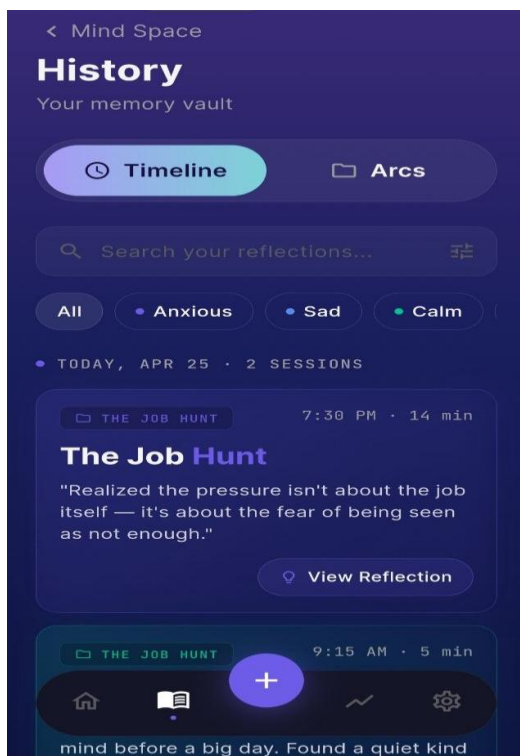


Figure 20- Timeline View

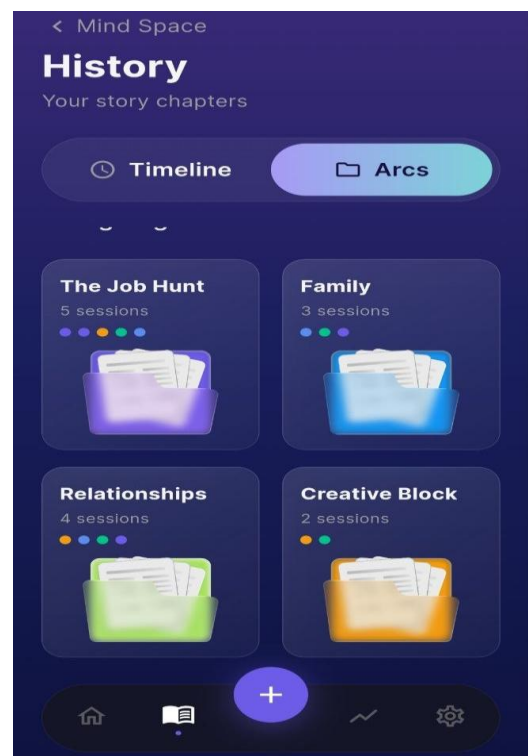


Figure 21 - History — Arcs Grid View

- **Arc Detail and Insight :**

The Arc Detail screen displays the emotional journey graph across sessions. Once an Arc reaches three sessions, the Generate Arc Analysis button becomes active and produces a longitudinal insight structured as emotional evolution and key pattern.



Figure 22 – Arc Detail — Emotion Graph

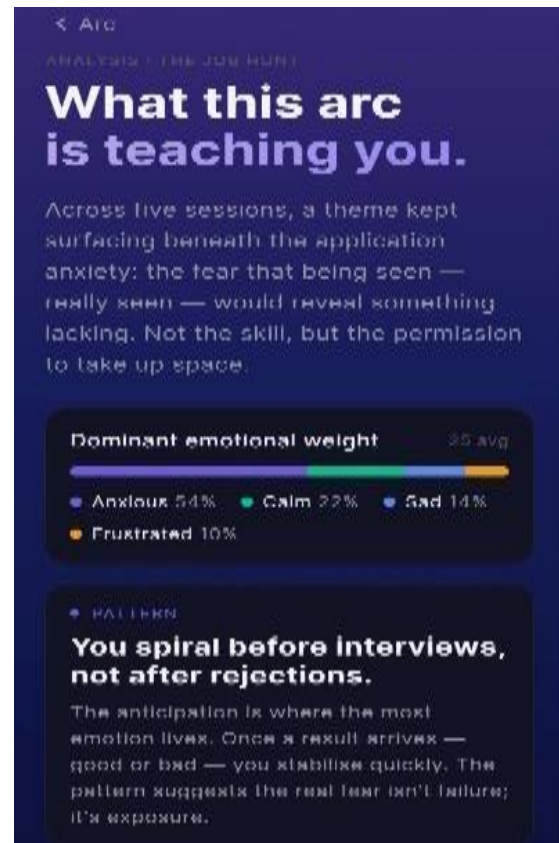


Figure 23 – Arc Detail — Generated Insight

- **Analysis and Profile :** The Analysis screen shows global statistics and a 14-week heatmap of reflection activity. The Profile screen provides personalisation options including Sage's conversational tone, colour palette, quiet hours, and data export.

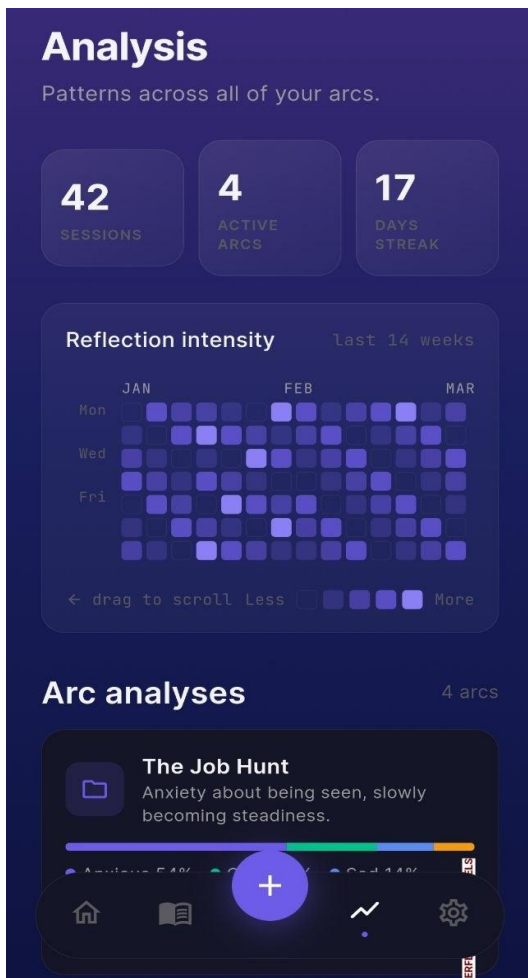


Figure 24 – Analysis screen

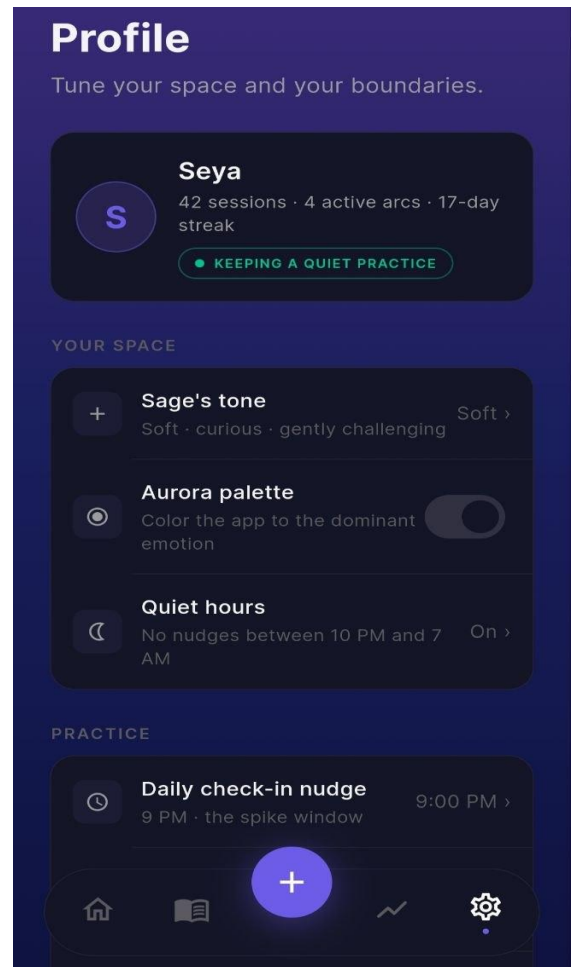


Figure 25 – Profile and Settings

3.4 Conclusion

This chapter presented the implementation of Mind Space, from the technology stack to the application interfaces. The system combines a Flutter frontend with a fully serverless AI backend, enabling streaming conversations, automatic Arc clustering via embedding similarity, and longitudinal insight generation all within a zero-cost infrastructure.

GENERAL CONCLUSION

This project was born from a genuine desire to help people understand themselves — not by diagnosing them, not by giving advice, but simply by listening and remembering. That desire became Mind Space.

Across three chapters, we moved from theory to reality. We began by identifying the limitations of existing journaling and mental health tools, and establishing the theoretical foundations of our approach. We then modelled the system using the RUP methodology, defining the core use cases and the interactions between the user, the administrator, and the application. Finally, we implemented a complete Android application built on Flutter and a fully serverless AI backend, bringing together three artificial intelligence services to power real-time conversation, safety classification, semantic embedding, and longitudinal insight generation.

The Arc auto-assignment algorithm stands as the core technical contribution of this work. By comparing reflection embeddings using cosine similarity, the system automatically groups conversations into meaningful emotional themes — called Arcs — without any manual input from the user. This is what separates Mind Space from a simple chatbot or journal.

Every design decision, from the dark interface to the way Sage responds without a bubble, was made with one goal: to create a space that feels quiet, private, and genuinely human.

This work has its limits. The evaluation remains small in scale, the application is Android-only, and multilingual support is documented as future work. But these limitations point clearly toward what comes next — collaboration with mental health professionals, expansion to iOS, and Arabic language support for our local context in Algeria.

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